

Working Principle of IAM:

IAM provides the infrastructure to manage access to your AWS resources. Now let's see how IAM works.

- **Principal** - An entity that can perform an action on AWS resources can be called a principal. Users who have access to AWS resources, an application that has access to AWS account, all are principals.
- **Request** - When a principal tries to perform an action on any resource then this process is called a request. Requests contain data like -
 1. Operations that a principal wants to perform.
 2. Resources on the operation which will be performed.
 3. Information regarding the principal.
- **Authentication** - For sending any request, a principal should be always authenticated. This means any principal should have a username and password for its unique identification. You can impose multi-factor authentication for some critical resources in your AWS account. Then some other information like secret key should be shared along with the username and password.
- **Authorisation** - To send any request, a principal must be authorised to do so. You can grant different level of access to different principals. AWS root user has access to all the resources by default.
- **Action** - Actions are the operations principals want to perform. After it is verified that principal is authentic and authorised, AWS approves the requested action.
- **Resources** - Principals perform actions using resources. It can be anything from EC2 instances, to S3 bucket, to an IAM user.

Overview of IAM users:

Using IAM policy you can impose customised security system in your AWS account. Different users can have different level of access. Broadly we can categorise users into three groups.

- **Root user** - By default, this user has access to every resource. This is the user you have created when creating the AWS account. This user creates the first IAM user.
- **IAM users** - These are the users who can access AWS resources depending on their rights.
- **Federating existing users** - You can give temporary access to the users who are already authenticated in your corporate network by federating existing identities into AWS.